

FLOW REGIMES IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

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- Flow regimes (magnitude, frequency, duration, timing, and rates of change in flow events) of all intermittent rivers and ephemeral streams (IRES) are characterized by variable periods of zero flow, caused by processes at different scales and often exacerbated by human activities
- When flow ceases, surface waters in many IRES dry to isolated pools but flow may continue through hyporheic sediments below the streambed. If drying continues, hyporheic flows may also cease
- Intermittence and drying have major consequences for aquatic ecosystem structure and functioning and so it is important to describe flow regimes of IRES. Data from gauging stations can be summarized into hydrological metrics (e.g., variance in frequency, duration, and timing of intermittence) and supplemented by wet-dry mapping, various forms of imagery, and modeling to classify riverine regimes
- These classifications show that IRES are globally abundant and that across much of the world, flow intermittence is increasing through climatic drying and greater rates of water extraction

2.2.1 INTRODUCTION

The flow regime of a river is defined as the temporal variability of its discharge, particularly the quantity, timing, and variability in flow (Poff et al., 1997; Nilsson and Renöfält, 2008; Sauquet et al., 2008). Flow regimes are generally expressed as statistical generalizations of hydrological phenomena (e.g., seasonal runoff patterns, median annual discharge, mean, and variance of peak flows) at a particular location over multiple years or decades (Thoms and Sheldon, 2000). Alternatively, flow regimes may be summarized as probability distribution functions of daily flows (e.g., Doyle et al., 2005) to provide information on average water availability, the extent of discharge fluctuations, and the frequency of high and low flows (Botter et al., 2013). Whatever form is used, information about flow regimes is crucial to most research programs and management strategies in rivers because of the central role played by discharge and its temporal variability in governing geomorphology, water quality, and ecology (Box 2.2.1). Extremes of the flow regime, including frequency, timing, and

BOX 2.2.1 FLOW REGIME FUNDAMENTALS

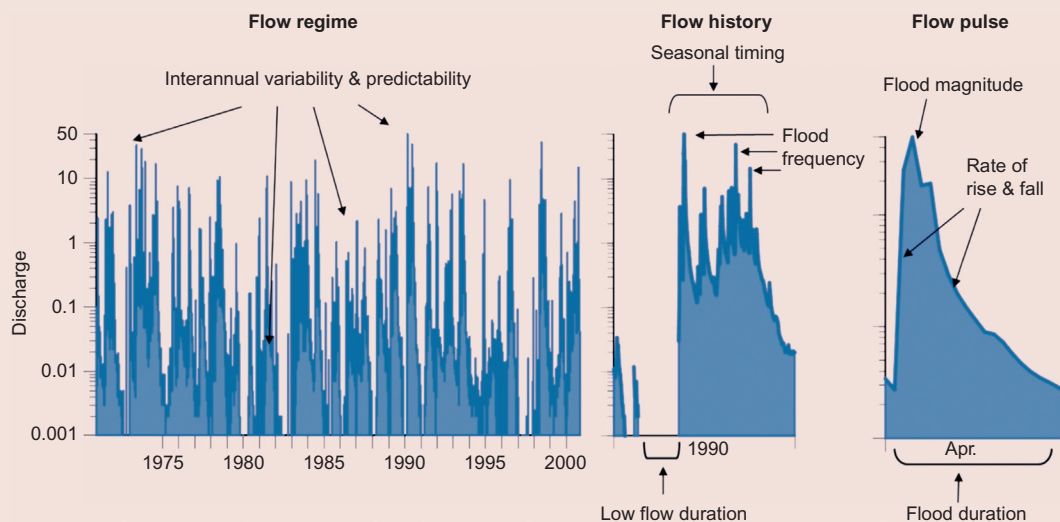
River flow regimes comprise five broad components: magnitude, frequency, duration, timing, and rate of change of hydrological conditions, as well as the temporal variability and predictability in these components. Magnitude refers to the amount of water at a given time interval and is usually expressed as discharge—the volume of water moving past a fixed point per unit time. Gauging stations typically record river height (stage) at regular time intervals and discharge is estimated from a rating curve that describes the relationship between river height and discharge at the gauging station (Gordon et al., 2004). Magnitude can also be described in relative terms (e.g., the amount of water that links disconnected pools along a river course). Changes in magnitude alter habitat types and availability in all IRES (Chapter 2.3).

Frequency refers to how often a flow above a given magnitude recurs over some specified time interval while duration is the period of time associated with a specific flow condition or event. Frequency and duration can also be described in relative terms; for example, the duration and frequency of a particular flow event that inundates a floodplain in a given area. Among the most important hydrological metrics in IRES are frequency and duration of periods of zero flow (e.g., Snelder et al., 2013) because these have major ecological implications for aquatic and terrestrial biota and ecosystem processes (Jaeger et al., 2014; chapters in Section 4).

Timing refers to when a particular event occurs or its regularity or predictability. Timing is ecologically important when a given flow event is needed at a certain time of the year (e.g., fish spawning) as well as being a useful indicator of the inherent variability of flows. The timing of intermittence can be as ecologically important as frequency and duration of intermittence (Leigh and Datry, 2016). Finally, rate of change (sometimes called “flashiness”) refers to how quickly flow changes from one magnitude to another; “flashy” streams have faster rates of change than “stable” streams.

These five components of flow regime can be readily illustrated on a hydrograph (Fig. 2.2.1). Varying the temporal scale of the hydrograph portrays the flow regime components in different ways: over multiple years to show interannual variability and predictability of all five components; over a single year to show durations of zero-flow periods and seasonal timing of floods; and over a single flow pulse to show flood magnitude and rates of change. Across all these temporal scales, these five components of the flow regime govern almost every physical, chemical, and biological aspect of all rivers, including IRES.

Alteration of flow regimes by human activities threatens the ecological integrity of river ecosystems and may result in serious declines in biodiversity and the provision of crucial ecosystem services. This “natural flow regime paradigm”, proposed in a seminal paper by Poff et al. (1997), is why river ecologists and water resource managers rely on detailed information about flow regimes: flow regime fundamentals.

**FIG. 2.2.1**

Components of the flow regime illustrated at three temporal scales.

Reproduced from Olden, J.D., Kennard, M.J., Pusey, B.J., 2012. A framework for hydrologic classification with a review of methodologies and applications in ecohydrology. Ecohydrology 5, 503–518, with permission.

duration of periods of flow cessation, are particularly important drivers in river ecosystems at multiple spatiotemporal scales (Walker et al., 1995; Larned et al., 2010b), yet their full effects remain poorly understood (Leigh et al., 2016).

In all intermittent rivers and ephemeral streams (hereafter, IRES), the flow regime is defined by periods of flow cessation at some time at one or more points along their network (Chapter 1). These periods vary widely, depending on regional and local factors (Bhamjee and Lindsay, 2011; McDonough et al., 2011; Costigan et al., 2016), and alter the extent, timing, and duration of hydrological connectivity in the longitudinal, lateral, and vertical dimensions of IRES (Chapter 2.3). All river networks include intermittent segments which may be concentrated in one part of the network (e.g., headwaters, lower reaches) or interspersed throughout the entire network (Lake, 2003; Turner and Richter, 2011; Chapter 2.3). In humid and temperate regions, IRES are often confined to the headwaters but in many arid regions, IRES account for most of the stream length and occur throughout the network (Caruso and Haynes, 2011).

As in perennial rivers, the flow regime of IRES governs channel form and sediment dynamics (Godsey and Kirchner, 2014; Chapter 2.1). Similarly, temporal fluctuations in water quality in many IRES correlate with variations in flow regime components (Chapters 3.1 and 3.2), especially timing and duration of intermittence (Sheldon and Fellows, 2010; Leigh, 2013; Costelloe and Russell, 2014). Although most research exploring associations between flow regimes and riverine ecology has focused on perennial rivers (Leigh et al., 2016), similar principles apply in IRES (Leigh and Sheldon, 2008; Datry et al., 2014; chapters in Section 4 of this book). For example, alteration of the natural flow regime (Poff et al., 1997, Box 2.2.1) typically reduces biodiversity of native aquatic species and impairs most riverine ecological processes (Bunn and Arthington, 2002; Rolls et al., 2012). These ecological impacts can be associated with specific aspects of the flow regimes of IRES, particularly changes to the timing, duration, and magnitudes of streamflow and intermittence (Leigh et al., 2012; Datry et al., 2016b; Leigh and Datry, 2016).

Another aspect to consider is that when flow ceases, surface water can persist in remnant pools that may be connected by hyporheic flow below the streambed (Boulton et al., 1992; Anna et al., 2009). In other cases, surface and hyporheic water quickly disappears (Datry, 2012; Vander Vorste et al., 2016a). The effects on biota and ecological processes differ depending on whether surface water remains or not (e.g., Anna et al., 2009; Corti et al., 2011). Traditional approaches to measuring flow regime (e.g., flow gauging) often cannot distinguish these two situations, promoting searches for alternative methods to describe IRES flow regimes. Finally, understanding the ecological effects of altered flow regimes in IRES helps water resource managers to protect and restore key components of flow regimes in IRES using tools such as environmental watering (Chapter 5.4) and “cease-to-pump” rules (Reinfelds et al., 2004).

This chapter reviews our hydrological knowledge about IRES flow regimes to provide context for understanding how intermittence creates temporal sequences of aquatic environments that fluctuate between flowing and nonflowing states, sometimes including complete loss of surface water. After summarizing factors that control the natural flow regimes of IRES, we outline various approaches such as wet-dry mapping, hydrological metrics, and modeling that have been used to describe IRES flow regimes. Case studies from different parts of the world illustrate the application and interpretation of these various approaches. We conclude by discussing knowledge gaps and likely directions for research into the flow regimes of IRES, especially how they might advance our understanding of the effects of altered flow regimes on geomorphology, water quality, biota, and ecological processes in IRES so that we might manage these systems more wisely.

2.2.2 CONTROLS ON THE NATURAL FLOW REGIME OF IRES

The natural flow regime refers to the characteristic pattern of a river's flow magnitude, duration, timing, frequency, and rate of change (Box 2.2.1). In IRES, intermittence is the defining feature of the natural flow regime and generates complex inter- and intraannual expansions and contractions of hydrological connectivity throughout the river network (Stanley et al., 1997; Costigan et al., 2016; Chapter 2.3). Flows in IRES may range from hours following precipitation events through to multiple months or even years. Much of the research on IRES flow regimes has focused on identifying physical controls of intermittence (e.g., Fritz et al., 2008) but many other environmental factors also appear relevant. These factors are reviewed by Costigan et al. (2016) across broad spatiotemporal scales, particularly the way that meteorological, geological, and land cover characteristics control the tendency of flow regimes toward intermittence (Fig. 2.2.2).

As with all rivers, the flow regime of any IRES reflects climatic and catchment processes that mediate how water enters and moves through the river network, interacts with local channel properties along the way. Flows occur when inputs such as rainfall or snowmelt exceed losses from infiltration and evapotranspiration or when water is released from existing storage (Godsey and Kirchner, 2014; Costigan et al., 2015). The interaction between catchment structure (e.g., topography, geology, vegetation) and climate is the primary control of all flow regimes (Jencso et al., 2009; Nippgen et al., 2011; Snelder et al., 2013) and occurs across spatiotemporal scales ranging from entire watersheds to microhabitats (<1 m) and from millennia to hours (Poff et al., 1997; Fig. 2.2.2). Some controls on streamflow can operate at small spatiotemporal scales (e.g., duration and intensity of local rainfall; Fleckenstein et al., 2006; Costigan et al., 2015) so it is important to look at the full range of spatiotemporal scales when understanding controls on intermittence (Costigan et al., 2016).

There have been very few investigations of the multiple mechanisms that control intermittence (Godsey and Kirchner, 2014), despite the long-recognized need for this understanding (Gomi et al., 2002). This paucity of research identifying how hydrological processes interact with geomorphic, climatic, and biological ones may arise from the challenges of integrating different research fields that adopt discipline-specific methods, philosophies, and spatiotemporal extents (e.g., Benda et al., 2002). Therefore, much of our understanding of the controls of stream flow in IRES has been inferred from perennial rivers (Costigan et al., 2016). There is clearly a need to focus on IRES, especially those subject to human activities that alter intermittence (Chapter 5.1). To guide this work, it is useful to review the methods currently used to characterize flow regimes in IRES.

2.2.3 METHODS TO CHARACTERIZE FLOW REGIMES OF IRES

A major challenge in characterizing IRES flow regimes is acquiring hydrological data because many IRES are ungauged, unmapped, or inaccurately depicted on topographic maps (Hansen, 2001; Caruso and Haynes, 2011). Precise spatial distributions of IRES are difficult to obtain (Benstead and Leigh, 2012; Datry et al., 2014), especially for intermittent headwaters routinely underestimated by traditional mapping techniques (Meyer and Wallace, 2001) and seldom gauged.

Where river networks are gauged, stations are usually sparsely distributed and tend to be on perennial rivers close to urban areas or places of economic interest. Headwater streams and other IRES are severely underrepresented (De Girolamo et al., 2015; Eng et al., 2016), leading to underestimation of

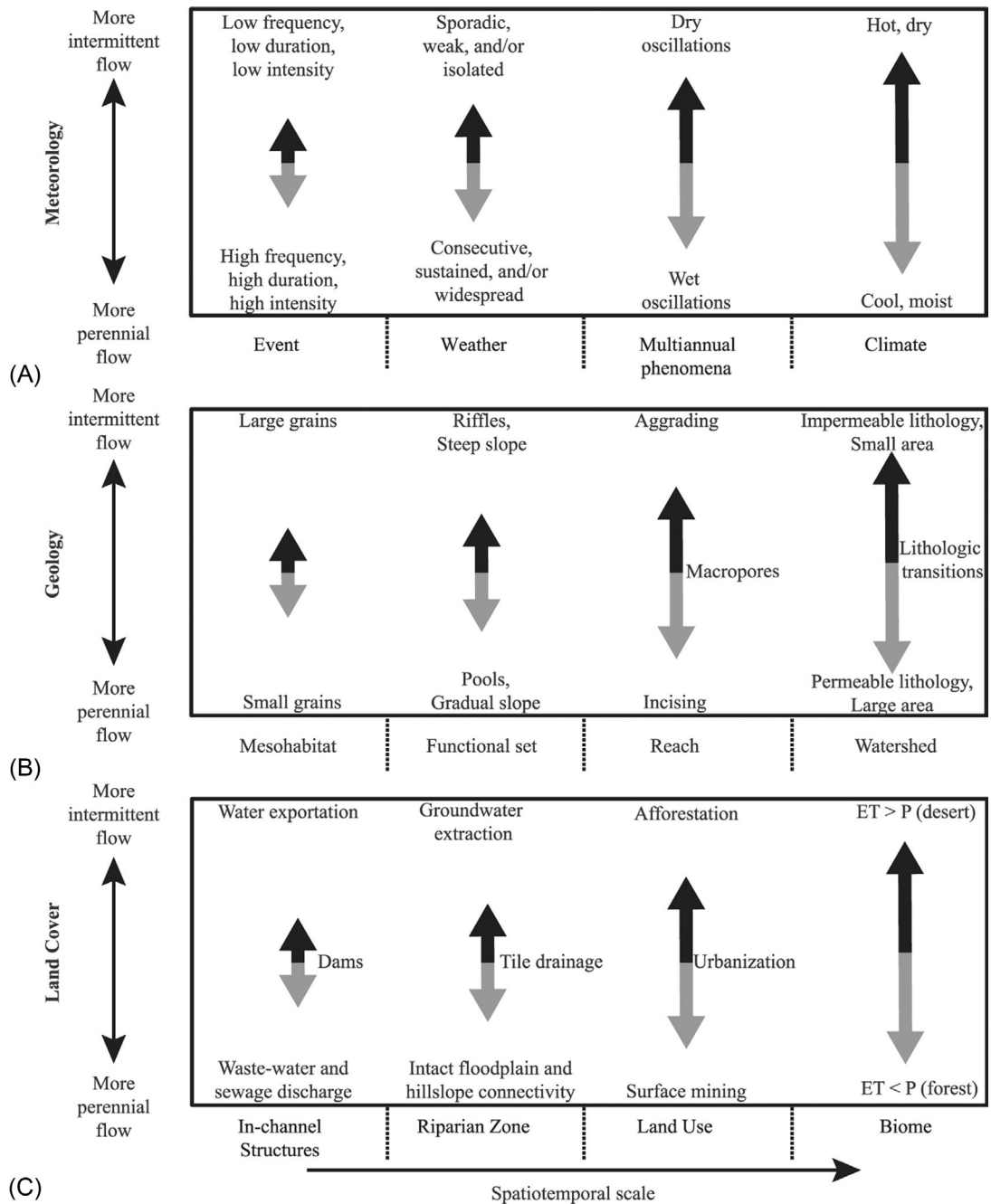


FIG. 2.2.2

Spatiotemporal hierarchy (x-axis) of meteorological (A), geological (B), and land-cover (C) controls on flow regime, with factors promoting either intermittence (*upper black arrows*) or perennial flow (*lower gray arrows*) along the y-axes. Alternative controls orthose with unknown directions of influence are presented near the origins of the arrows.

Reproduced from Costigan, K.H., Jaeger, K.L., Goss, C.W., Fritz, K.M., Goebel, P.C., 2016. Understanding controls on flow permanence in intermittent rivers to aid ecological research: integrating meteorology, geology and land cover. Ecohydrology doi: 10.1002/eco.1712, with permission.

the regional extent and distribution of flow intermittence (Snelder et al., 2013). Extrapolation of gauge data to characterize patterns across the entire network is perilous because the point measurements at the gauging stations seldom represent spatiotemporal variability in ungauged locations, particularly in IRES in arid and semiarid areas. Added to this is the notoriously variable quality and quantity of discharge data from most gauging stations (Kennard et al., 2010b). Finally, to adequately assess the inherently greater variability of hydrological data from most IRES, long time series (>15–30 years) of streamflow data are desirable (Kennard et al., 2010b; Snelder et al., 2013).

Therefore, alternative approaches to using gauged data to characterize IRES flow regimes are increasing in popularity and technical quality (Datry et al., 2016a; Gallart et al., 2016). These approaches include wet-dry mapping and recent advances in various imaging and modeling techniques. Data from these approaches can be used to supplement traditional sources such as gauged discharge data and often provide valuable extra information such as the duration of isolated pool phases which are of great ecological significance to IRES biota (Boulton, 1989; Rossouw et al., 2005).

WET/DRY MAPPING

On-ground mapping to track temporal changes in the distribution of surface water in IRES networks (e.g., Stanley et al., 1997) has provided useful insights into altered hydrological connectivity (Chapter 2.3) and spatial variability in flow regimes along IRES. Although wet/dry mapping generates valuable data, it demands considerable time in the field, especially when the mapping is conducted at frequent intervals and over large areas. To address this intense demand on field time, citizen scientists have been recruited to assist in wet/dry mapping of IRES (Turner and Richter, 2011; Datry et al., 2016a; Fig. 2.2.3).

Properly trained citizen scientists have proven to be as effective as experts for tasks such as mapping the extent of reaches with flow, isolated pools, and dry channels along IRES (Turner and Richter,



FIG. 2.2.3

Field surveys of flow regime and water presence in IRES demand considerable time and personnel, two issues that can be addressed by involving citizen scientists in the work.

Photo: Courtesy A. Boulton.

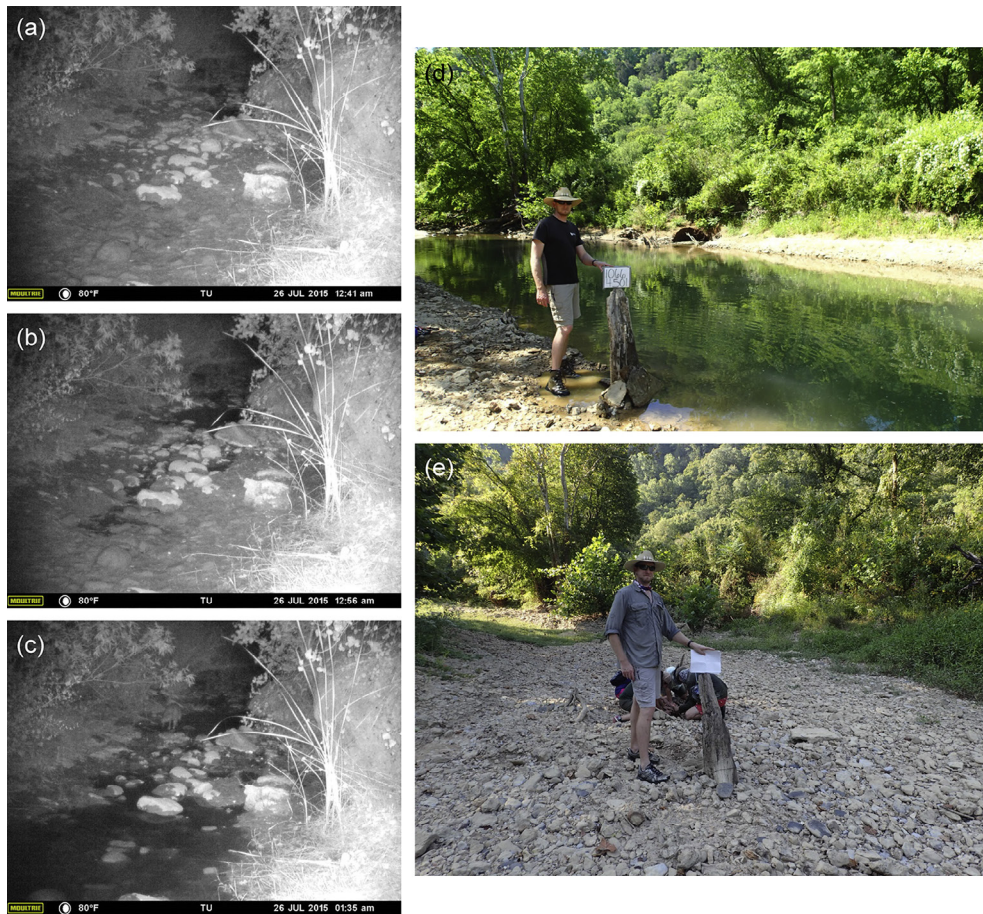
2011), and these studies are especially useful for the inexpensive collection of long-term data. One example is the 12 years of wet/dry mapping carried out along an 80-km segment of the San Pedro River (Arizona, USA) described by Turner and Richter (2011). A second example is the visual mapping of some 4000 km of wet and dry river reaches of over 20 IRES networks in Western France by trained volunteers every 15 days during summer that has been done since 2009 (Datry et al., 2016a). This has yielded catchment-scale maps illustrating temporal changes in flow regime at multiple points along different IRES and changes in surface-water connectivity that lead to habitat fragmentation (Chapter 4.9).

Data collected by citizen scientists help compensate for the limited number of gauging stations along many IRES and have improved our ability to characterize IRES flow regimes at specific locations in the network. Coupled with data from existing gauging stations or groundwater monitoring bores nearby, wet/dry mapping can be used to describe, model, and predict the typically complex and dynamic spatiotemporal flow patterns in IRES (Chapter 2.3). Wet-dry mapping is particularly valuable when qualitative states of IRES flow regimes are described such as the six “critical steps” in a drying sequence (Boulton, 2003), the five “hydrological conditions” (Fritz et al., 2006), or the six “aquatic states” (Gallart et al., 2012) simplified to three by Gallart et al. (2016) and used by Datry et al. (2016a). With appropriate training and coordination in river mapping, citizen scientists could monitor real-time flow patterns and qualitative states of IRES flow regimes in nearby IRES, aided by the increasing availability of smartphones with cameras, GPS devices, and high-speed internet connections.

IMAGERY: FROM SATELLITES TO SITE CAMERAS

Various remote sensing techniques have been used to measure proxies for discharge (e.g., river height or width) from which to estimate flow regimes. These techniques include air- and space-borne measurements of surface velocity; radar altimeters to measure surface-water elevations; space-borne measurements of gravity fluctuations to estimate river discharge; and measurements of wetted areas and bank heights to estimate flow volumes, hydrological persistence, and flood extents (e.g., Puckridge et al., 2000; Costa et al., 2013; Gleason and Smith, 2014). Remotely sensed data provide effective proxies for flows in large rivers and their floodplains but are unsuitable for small channels such as those of headwater streams because of resolution issues and the density of vegetation cover (Benstead and Leigh, 2012). As most headwater streams are intermittent, this limits the application of satellite imagery for mapping many IRES. Although techniques are being increasingly refined for integrating very-high-spatial-resolution multispectral remote sensing with updated algorithms to map ephemeral stream networks (e.g., Hamada et al., 2016), imagery using light aircraft, drones, and mounted cameras is more practical in smaller streams.

Repeated imagery (e.g., time-lapse photography) can be used to characterize temporal variation in the presence or absence of surface water in sections of smaller IRES, enabling derivation of flow regimes as changes in stream level over time (Puckridge et al., 2000). Fixed cameras, often accompanied by stream gauging instrumentation, can be deployed for long periods and programmed to take photographs at specified intervals (Fig. 2.2.4a-c) or when initiated by movement or flow sensors. In larger rivers, the combined use of multitemporal satellite data with flow gauge and groundwater level data can provide an efficient tool to characterize and model flow regimes (Costa et al., 2013). In smaller rivers, repeated photographs at a single location (Fig. 2.2.4d and e) are visually powerful ways to portray the flow regime of IRES at different times of the year. These repeated images can then be downloaded on websites to reveal long-term variations in stream flow and channel morphology (e.g., http://www.konza.k-state.edu/keep/sci_adventures/streams/oxbow/oxbow_gallery.htm;

**FIG. 2.2.4**

(a-c) Time-lapse images of a reach in Salome Creek, Arizona, where the stream dries during the day but water reappears at night, providing a resource for local wildlife. Repeated camera images from consistent photo points, such as these paired images (d, e) from Spring Creek, Tennessee, are a powerful way to illustrate seasonal variation in flow in IRES.

Photos: Courtesy M. Smart (a-c), A. Doll (d), W. Curtis (e).

<http://datawarehouse.wrl.unsw.edu.au/maulescreek/eastlynn/>). Recent developments in airborne technology and remote camera image processing represent promising advances toward better mapping and understanding of IRES and their flow regimes.

FIELD LOGGERS AND FLOW SURROGATES

Various types of data loggers are commonly used to characterize IRES flow regimes when flow gauges are absent. These loggers measure flow surrogates such as electrical conductivity (e.g., Gungle, 2007; Chapin et al., 2014), water temperature (e.g., Constantz et al., 2001; Gungle, 2007), and water level and/or the

presence-absence of water (e.g., Costelloe et al., 2005; Bhamjee and Lindsay, 2011; Vander Vorste et al., 2016b). Time series data from the loggers can be used to track the movement of wetting and drying fronts (Bhamjee and Lindsay, 2011) and the persistence of surface waters in different reaches (Vander Vorste et al., 2016b). These data can then be translated into hydrological metrics to assess and compare flow regimes (*Hydrological Metrics* section of this chapter). For example, time series data of water presence and absence provided by water-state loggers generate point estimates of the duration and frequency of drying that can then be matched with ecological changes in IRES (e.g., Jaeger et al., 2014; Vander Vorste et al., 2016b).

Thanks to their relatively low prices (Chapin et al., 2014), arrays of multiple loggers can be placed within and across IRES networks to spatially describe flow and water presence-absence regimes and connectivity in space and time (Jaeger and Olden, 2012; Vander Vorste et al., 2016b). These arrays are especially useful if the study areas are remote and frequent visitation is impractical. However, there are several limitations to using loggers. For example, in-stream and riparian loggers may be washed out or buried during flood events and subject to vandalism. Therefore, regular maintenance and data downloads are required, particularly where flow regimes are flashy or risks of vandalism are high. Most sensors are unable to differentiate flowing or standing waters. Sometimes detecting the presence of water is difficult; if moist sediment builds up on the sensor probes, the sensors interpret that as wet conditions. Regular site visits are required to ground truth the sensor datasets.

HYDROLOGICAL METRICS

Hydrological metrics are indices or statistics computed from multiyear time series of discharge data and used to characterize statistical properties of various components of riverine flow regimes. They have been widely used to evaluate spatial variation among flow regimes (e.g., Puckridge et al., 1998; McMahon et al., 2007), classify rivers (e.g., Hughes and James, 1989; Poff, 1996), and explore associations between flow regimes and biotic structure or ecological processes (e.g., Kennard et al., 2007; Leigh, 2013). Despite the array of hydrological metrics available to characterize river flow regimes (reviewed by Olden et al., 2012), relatively few are employed to characterize intermittent flows. The principal ones are number of zero-flow days, number of zero-flow periods, average duration of zero-flow periods, percent flow permanence in a year, and percent flow permanence by season (e.g., Snelder et al., 2013; Jaeger et al., 2014; Costigan et al., 2015; Schriever et al., 2015).

In an example illustrating the capacity of hydrological metrics to discriminate different flow regimes, including those of IRES, Kennard et al. (2010a) used 120 metrics to classify discharge data from 830 stream gauges across Australia. Eight of the 12 classes of rivers in this study were classified as intermittent: 4 classes of rivers that rarely ceased flow, 3 that regularly ceased flow, and 1 that was extremely intermittent. The hydrological metrics that split perennial rivers from IRES included the number of zero-flow days, baseflow contributions, and magnitude of daily runoff (Fig. 2.2.5), indicating that not just flow intermittence but other (nonzero) flow conditions are important in characterizing IRES flow regimes. There was often greater variance (expressed by the larger boxes and longer “whiskers”) of these hydrological metrics within the eight classes of IRES compared to the four perennial classes (Fig. 2.2.5); this pattern of greater variance for IRES was also evident for many of the other metrics (Kennard et al., 2010a).

Suites of hydrological metrics have also been used to identify distinct groups of flow regimes of IRES in other parts of the world. Using 10 hydrological metrics to classify long-term (15–58 years) daily streamflow data from 806 relatively undisturbed streams in the United States, Poff (1996) discerned three classes of IRES from seven perennial-stream classes. Again, number of zero-flow days was a key metric, along with higher unpredictability and coefficients of variation for flows measured at different temporal

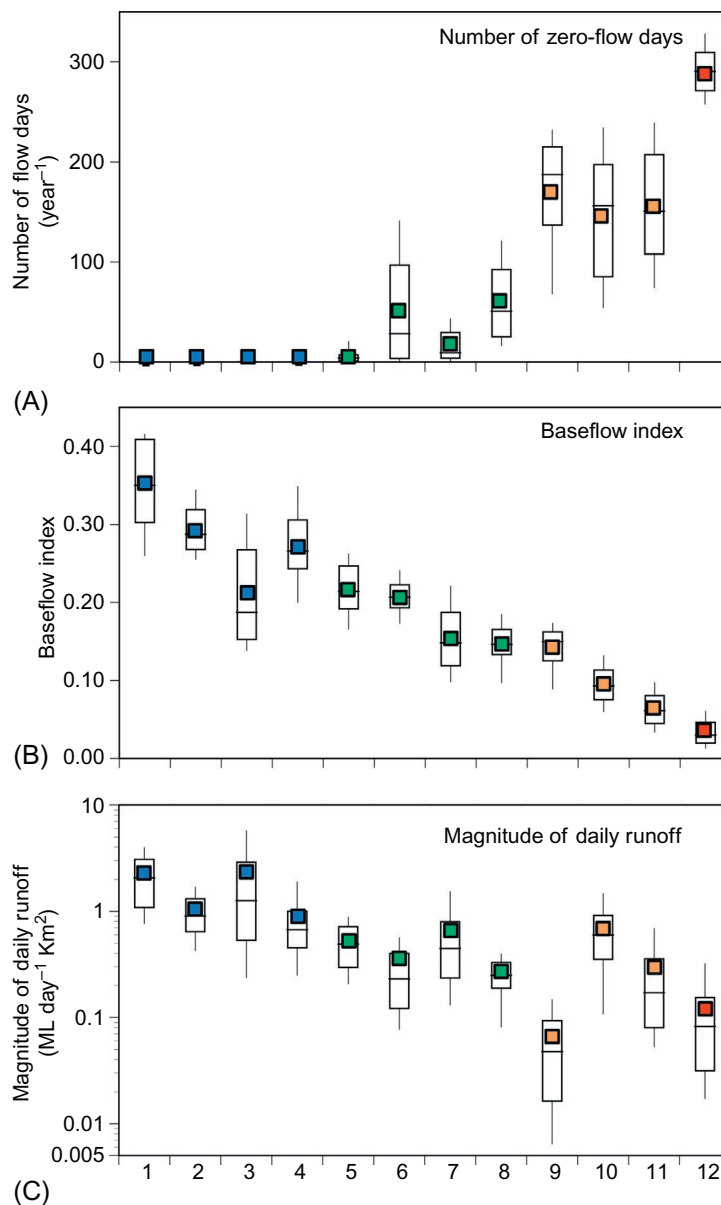


FIG. 2.2.5

Box plots of number of zero-flow days (A), baseflow index (B), and magnitude of daily runoff (C) in 12 classes from a classification of flow regimes of Australian rivers: 1, stable baseflow; 2, stable winter baseflow; 3, stable summer baseflow; 4, unpredictable baseflow; 5, unpredictable winter rarely intermittent; 6, predictable winter intermittent; 7, unpredictable intermittent; 8, unpredictable winter intermittent; 9, predictable winter highly intermittent; 10, predictable summer highly intermittent; 11, unpredictable summer highly intermittent; 12, variable summer extremely intermittent. Rivers in classes 1–4 (blue boxes) have primarily perennial flow; those in classes 5–8 rarely cease flow; those in classes 9–11 regularly cease flow; and class 12 rivers are extremely intermittent. Colored symbols are means; horizontal lines at the top, middle, and bottom of each open box represent the 75th, 50th (median), and 25th percentiles, respectively, and vertical bars (whiskers) represent 90th and 10th percentiles.

Modified from Kennard, M.J., Pusey, B.J., Olden, J.D., Mackay, S.J., Stein, J.L., Marsh, N., 2010a. Classification of natural flow regimes in Australia to support environmental flow management. *Freshw. Biol.* 55, 171–193; Kennard, M.J., Mackay, S.J., Pusey, B.J., Olden, J.D., Marsh, N., 2010b. Quantifying uncertainty in estimation of hydrologic metrics for ecohydrological studies.

River Res. Appl. 26, 137–156, with permission.

scales (days to years). Worldwide, studies of regional flow regimes have used number or frequency of zero-flow days as the primary hydrological metric to define or discriminate classes of intermittence (e.g., South Africa—[Smakhtin and Toulouse, 1998](#); France—[Snelder et al., 2013](#); European Mediterranean region—[Gallart et al., 2012](#); United States—[Reynolds et al., 2015](#); [Eng et al., 2016](#)). Consequently, it is logical to focus analyses and comparisons of the flow regimes of IRES on hydrological metrics related to zero flows, arrayed in [Table 2.2.1](#) according to the five main components of frequency, duration, timing and seasonality, and rate of change in flow events before and after periods of zero flows.

Table 2.2.1 Candidate hydrological metrics used to characterize zero-flow conditions in IRES flow regimes

Hydrological metric	Definition
<i>Frequency of flow events—zero-flow conditions</i>	
Zero-flow spell count ^a	Mean number of annual, seasonal, or monthly occurrences during which the magnitude of flow remains at or below some threshold defined as zero flow
CV zero-flow spell count ^a	Coefficient of annual, seasonal, or monthly occurrences during which the magnitude of flow remains at or below some threshold defined as zero flow
<i>Duration of flow events—zero-flow conditions</i>	
Zero-flow spell duration ^a	Mean duration of annual, seasonal, or monthly occurrences during which the magnitude of flow remains at or below some threshold defined as zero flow
CV zero-flow spell duration ^a	Coefficient of variation in duration of annual, seasonal, or monthly occurrences during which the magnitude of flow remains at or below some threshold defined as zero flow
Number of zero-flow days	Mean annual number of days having a magnitude of flow at or below some threshold defined as zero flow
CV number of zero-flow days	Coefficient of variation in annual number of days having a magnitude of flow at or below some threshold defined as zero flow
<i>Timing and seasonality of flow events—zero-flow conditions</i>	
Julian date of annual zero flow	The mean Julian date of the 1-day annual zero flow over all years
CV Julian date of annual zero flow	Coefficient of variation in Julian date of the 1-day annual zero flow over all years
Six-month seasonal predictability of zero-flow periods	Multiannual frequencies of zero-flow months for the contiguous 6 wetter months of the year divided by the multiannual frequencies of zero-flow months for the remaining 6 drier months. Wet and dry 6-month periods are those with fewer and more zero-flow frequencies, respectively
Predictability (P) of zero-flow days ^b	Colwell's (1974) predictability (P) of zero-flow days
Seasonality (M/P) of zero-flow days ^b	Colwell's (1974) seasonality (M/P) of zero-flow days
<i>Rate of change in flow events—before/after zero-flow spell^a</i>	
Rise rate	Mean rate of increases in flow magnitude (rising limb of hydrograph) over a given time period
CV rise rate	Coefficient of variation in rate of increases in flow magnitude over a given time period

Continued

Table 2.2.1 Candidate hydrological metrics used to characterize zero-flow conditions in IRES flow regimes—cont'd

Hydrological metric	Definition
Fall rate	Mean rate of decreases in flow magnitude (falling limb of hydrograph) over a given time period
CV fall rate	Coefficient of variation in rate of decreases in flow magnitude over a given time period
Number of reversals	Number of increases then decreases in flow magnitude over a given time period
CV reversals	Coefficient of variation in number of increases then decreases in flow magnitude over a given time period
CV, coefficient of variation.	
^a Independence criteria for zero-flow spell frequency and duration need to be specified (e.g., at least 7 days between spells).	
^b Colwell's (1974) predictability (P) of flow is composed of two independent, additive components: constancy (C—a measure of temporal invariance) and contingency (M—a measure of periodicity).	
Compiled or adapted from Olden, J.D., Poff, N.L., 2003. Redundancy and the choice of hydrologic indices for characterizing streamflow regimes. <i>River Res. Appl.</i> 19, 101–121; Kennard, M.J., Mackay, S.J., Pusey, B.J., Olden, J.D., Marsh, N., 2010b. Quantifying uncertainty in estimation of hydrologic metrics for ecohydrological studies. <i>River Res. Appl.</i> 26, 137–156; and Gallart, F., Prat, N., García-Roger, E.M., Latrón, J., Rieradevall, M., Llorens, P., et al., 2012. A novel approach to analysing the regimes of temporary streams in relation to their controls on the composition and structure of aquatic biota. <i>Hydrol. Earth System Sci.</i> 16, 3165–3182 and supplemented with some proposed metrics associated with rates of rise and fall in flow.	

The frequency and duration of zero-flow periods are particularly useful in characterizing intermittency (Poff, 1996; Knighton and Nanson, 2001) and remain the most common indices used in describing intermittence (e.g., Larned et al., 2010b; Costigan et al., 2015; Reynolds et al., 2015). These two metrics are also particularly relevant because of their direct translation to habitat availability and persistence, so they are increasingly reported in ecological studies of IRES (Jaeger et al., 2014; Schriever et al., 2015; Chapter 4.9).

MODELING

When flow gauges have only limited data for a location of interest, modeling approaches can be used to estimate data from which metrics can be calculated. Such models are also employed to simulate flows under changing environmental conditions, such as global change and water extraction for human uses (Chapter 5.1). Hydrologists often distinguish physical or deterministic models that typically predict time series of simulated daily flows (and from which metrics can be calculated) from statistical models used to directly estimate flow metrics (e.g., Carlisle et al., 2010).

However, most models for deriving flow regimes have been developed for perennial rivers and there remain serious limitations in accurately simulating hydrological extremes, including low-flow events that may lead to zero flows. There are various reasons for these difficulties in simulating zero-flow conditions. For example, most rainfall-runoff models concentrate on representing surface water processes (e.g., Pilgrim et al., 1988) and fail to adequately capture the effects of surface-groundwater interactions or local geological peculiarities such as karstic areas on river flow regime. Poor performances also result from complex nonlinear processes in runoff generation in low-yielding basins (Ye et al., 1997), from inadequate representation of spatial variation in the catchment characteristics that determine runoff or storage (Costelloe et al., 2005) and/or from an unsuitable formulation of the recession phase that does not allow streams to completely dry (Ivkovic et al., 2014).

To address some of these limitations, [Ivkovic et al. \(2014\)](#) developed a modified version of the IHACRES (Identification of Hydrographs And Components from Rainfall, Evaporation, and Streamflow data) model which is a hybrid conceptual-metric model that uses metric approaches to reduce inherent parameter uncertainty while simultaneously attempting to represent more detail of the internal processes than typical metric models ([Croke and Jakeman, 2007](#)). This modified version includes explicit representation of groundwater storage and a linear relation between storage and baseflow that replaces the classical slow transfer function. [Ivkovic et al. \(2014\)](#) applied this modified version to modeling the flow regime in an ephemeral river system in New South Wales, Australia, and found that it successfully simulated the transitions between baseflows and zero-flow periods within the ephemeral stream.

A second example of hydrological modeling of flow regimes in IRES is the work by [Cipriani et al. \(2014\)](#) who applied a postprocessing technique, based on a quantile-mapping approach (e.g., [Snover et al., 2003](#)), to the outputs of two rainfall-runoff models to simulate zero-flow events. The postprocessing technique is a convenient way to simulate zero-flow events without modifying the structures of the models. Application of this approach to a river in southeastern France that ceased to flow due to losses toward the underlying aquifer yielded a good match between the simulated and bias-corrected discharge time series from two rainfall-runoff models.

A third modeling approach is the Soil Water Assessment Tool (SWAT, [Gassman et al., 2007](#)), a public-domain program that has become one of the most widely used catchment-scale hydrological transport models in the world. The hydrological component of the model accounts for precipitation, evapotranspiration, surface runoff, infiltration, lateral flow, and percolation and has been successful in modeling intermittence in IRES ([Chahinian et al., 2011](#); [De Girolamo et al., 2016](#); [Tzoraki et al., 2016](#)), with a >85% match between modeling projections and real-world conditions. The SWAT model has also been effective in projecting IRES flow regimes under natural, current, and future conditions ([Jaeger et al., 2014](#); [Tzoraki et al., 2016](#)). Finally, random forest modeling approaches use models that are first trained on gauges with environmental data and then extrapolated to areas without gauges. These models rely upon conditional interference trees that assess relationships between environmental variables and flow metrics. Random forest modeling explained up to 83%, but typically 45%–50%, of the variance in the streamflow metrics of ungauged portions of watersheds within the Upper Colorado River basin ([Reynolds et al., 2015](#); [Reynolds and Shafroth, 2016](#)).

However, severe limitations persist in using models to derive IRES flow regimes. All models, whether deterministic, statistical, or a combination, are constrained by the amount and quality of available training data and only perform as well as our understanding of processes allows. In most cases, we lack adequate understanding of the different processes driving intermittence in different IRES ([Smakhtin, 2001](#); [Snelder et al., 2013](#); [Section 2.2.2](#)) and seldom have adequate data to train models to accurately predict flow regimes of any river type, especially IRES. Major advances in modeling flow regimes in IRES are needed ([Section 2.2.5](#)), supplemented with better datasets from IRES around the world.

2.2.4 DESCRIBING AND CLASSIFYING FLOW REGIMES OF IRES: CASE STUDIES

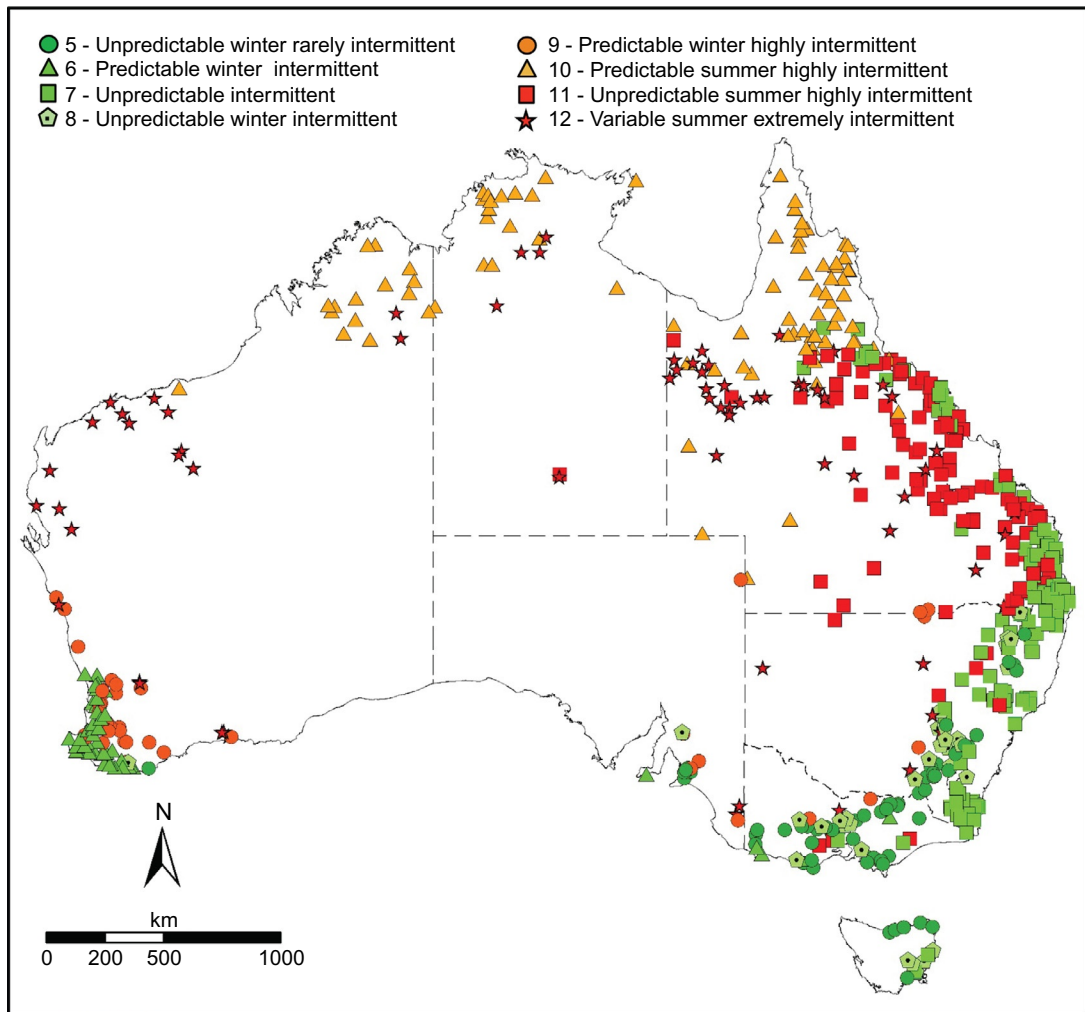
To illustrate applications and results of methods described in the previous section, we present several case studies of their use in IRES. In particular, we wanted to illustrate challenges and approaches to obtaining suitable flow-regime data, various ways of illustrating flow-regime data, and how these data can be used to draw hydrological conclusions about IRES in different parts of the world.

As mentioned in [Section 2.2.3](#), [Kennard et al. \(2010a\)](#) used multiple hydrological metrics to classify flow regimes in Australian rivers but there were some major challenges in obtaining an adequate dataset. As the study was conducted at a continental scale, the strategy for data collection needed to maximize the number and spatial coverage of gauges for analysis while ensuring they had comparable data quality and quantity. The initial dataset comprised mean daily discharge data for 2686 gauges with >10 years of records. Data from these 2686 gauges were then screened to provide a subset that (i) had few or no hydrological modifications caused by human activities; (ii) spanned at least 15 years from Jan. 1, 1965 to Dec. 31, 2000, preferably extending throughout a common year (i.e., 1980); and (iii) comprised continuous mean daily discharge data where possible. The first criterion reflected the study's aim to classify "natural" flow regimes. The second criterion was based on analyses of data adequacy by [Kennard et al. \(2010b\)](#) which also stipulated the need for the discharge records to lie within a discrete temporal window sharing >50% overlap among records. The third criterion also related to data adequacy by seeking the most complete records possible; where fewer than 30 days of data were missing, the missing records were imputed using linear interpolation or regression.

This screening process yielded a final dataset of 830 gauges covering most of the climatic, and hence potential flow regime, types across Australia ([Kennard et al., 2010a](#)) and sharing comparable lengths and timing of discharge records ([Kennard et al., 2010b](#)). Hydrological classification was undertaken using Bayesian mixture modeling which models the observed distribution of data as a mixture of a finite number of component distributions to determine the number of distributions, their parameters, and object memberships ([Webb et al., 2007](#)). The approach is fully probabilistic and uncertainty can be explicitly represented in terms of data specification (in this case, hydrological metric estimation uncertainty), class specification, and the final classification chosen. The most likely classification from the Bayesian analysis generated 12 classes of distinct flow regime types, eight of which featured intermittence. The geographical distribution of these eight classes varied in spatial cohesion ([Fig. 2.2.6](#)); the two classes described as "predictable winter highly intermittent" and "variable summer highly intermittent" were particularly scattered across the continent whereas other classes such as "predictable winter intermittent" had most of their representative gauges in the same part of the continent. In addition to geographic factors, climatic and several topographic factors were also generally strong discriminators of flow-regime classes ([Kennard et al., 2010a](#)). Major differences among the various classes of IRES related to their degree of intermittence (as number and frequency of zero-flow days) and the predictability and seasonal timing of flow.

Another study that classified rivers by flow regime is that by [Snelder et al. \(2013\)](#) of rivers in France and is noteworthy because it focused specifically on intermittence. Again, there was an extensive data screening process to derive a suitable dataset. The initial dataset comprised records of daily mean flow from over 3800 gauging stations across France. The first screening step removed poor quality data and records from gauges whose flows were likely to be modified by reservoirs, diversions, or significant abstractions upstream. The second screening step was to select data from the remaining stations that spanned the 35-year period from 1975 to 2009. Finally, these records were scanned for gaps longer than 20 days; where these gaps were found, the year of record was removed to yield a final dataset of 628 stations with 23–35 years of record.

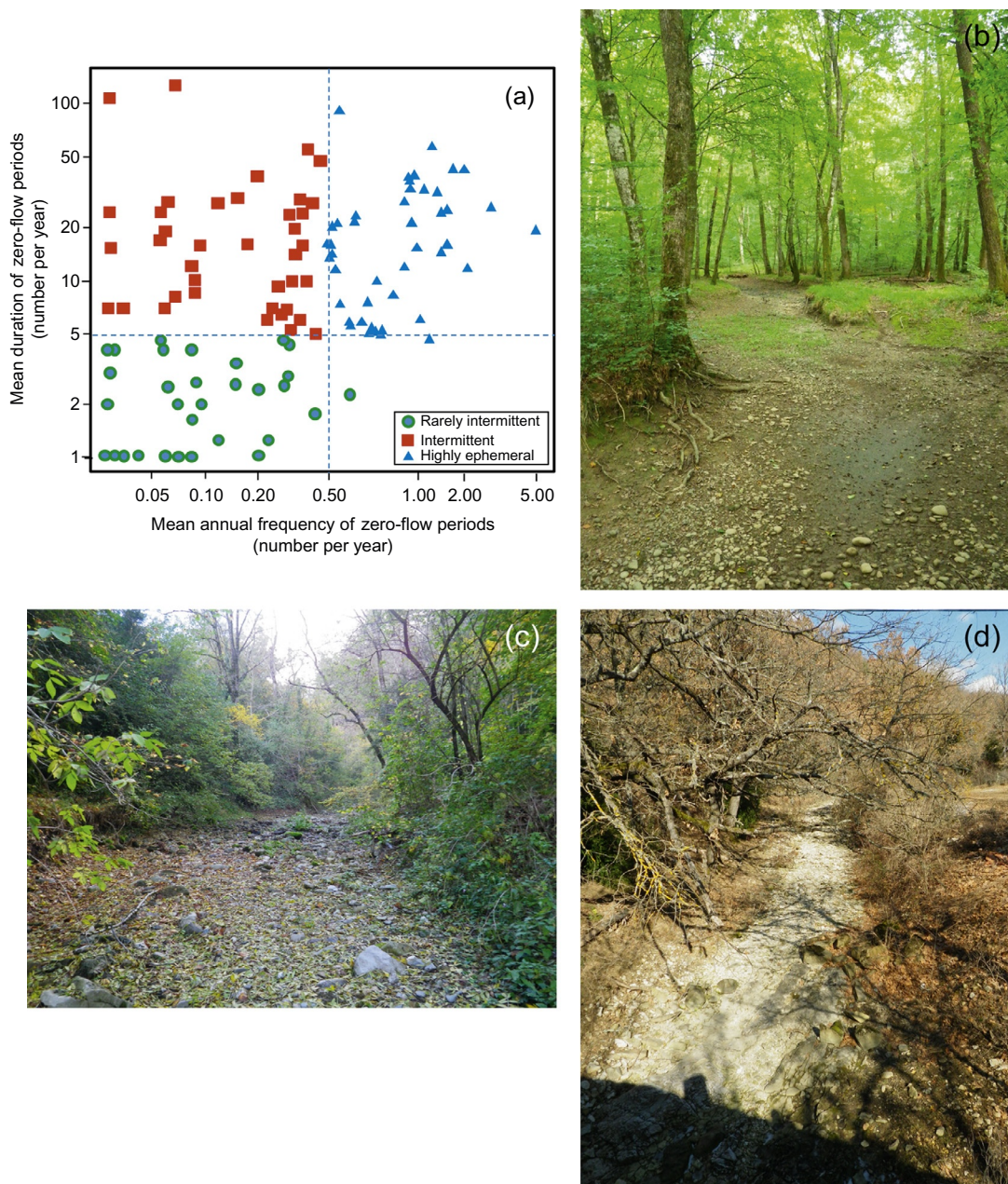
Daily flow data from each gauging station were used to produce two sets of time series describing flow intermittence in each calendar year of record: the frequency of zero-flow periods (consecutive days of zero flow) and the total number of zero-flow days. Two further hydrological metrics were generated from these for each station: the mean annual frequency of zero-flow periods and the mean

**FIG. 2.2.6**

Distribution of Australian rivers in eight classes of IRES (5–12, see Fig. 2.2.5) based on the analyses by Kennard et al. (2010a).

Modified from Kennard, M.J., Pusey, B.J., Olden, J.D., Mackay, S.J., Stein, J.L., Marsh, N., 2010a. Classification of natural flow regimes in Australia to support environmental flow management. *Freshw. Biol.* 55, 171–193; Kennard, M.J., Mackay, S.J., Pusey, B.J., Olden, J.D., Marsh, N., 2010b. Quantifying uncertainty in estimation of hydrologic metrics for ecohydrological studies. *River Res. Appl.* 26, 137–156, with permission.

duration of zero-flow periods (Snelder et al., 2013). These metrics were then used to classify the flow regimes from the 628 gauging stations into four classes: a single perennial-stream class whose regimes lacked zero-flow days, and three subclasses of IRES whose boundaries were based on a scatterplot of mean annual frequency of zero-flow periods versus mean duration of zero-flow periods (Fig. 2.2.7a).

**FIG. 2.2.7**

(a) Three classes (colored symbols) of IRES in continental France, grouped by their mean duration and mean annual frequency of zero-flow periods derived from 23 to 35 years of record between 1975 and 2009. Examples of the classes are la Clauge, rarely intermittent (b); la Vis, intermittent (c); and les Valsaintes, highly ephemeral (d).

Plot modified from Snelder, T.H., Datry, T., Lamouroux, N., Larned, S.T., Sauquet, E., Pella, H., et al., 2013. Regionalization of patterns of flow intermittence from gauging station records. *Hydrol. Earth Syst. Sci.* 17, 2685–2699, with permission. Photos: Courtesy B. Launay.

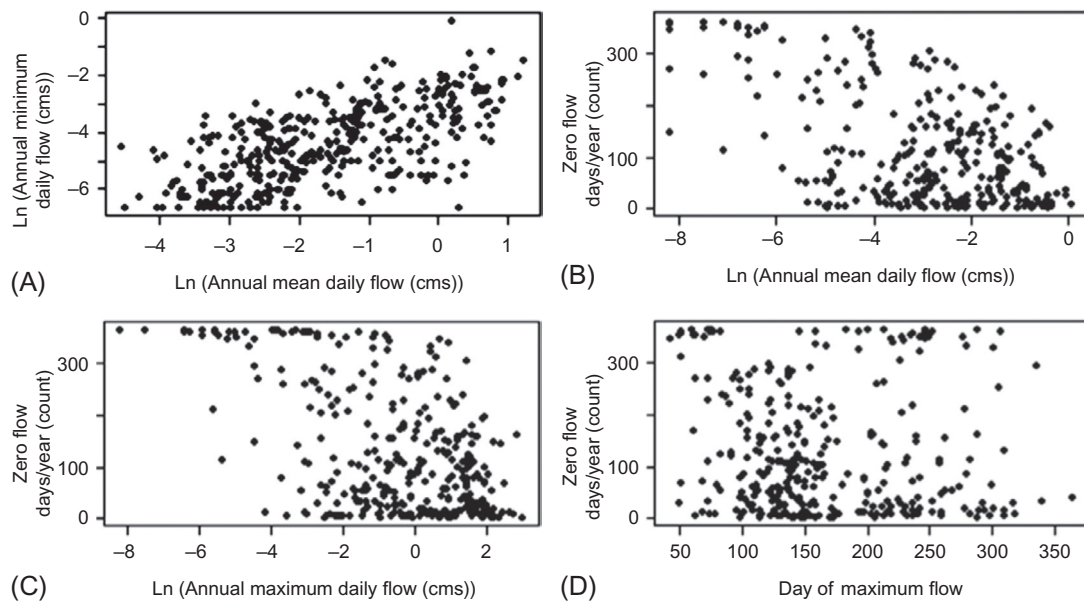
The three subclasses of IRES (examples in Fig. 2.2.7b–d) can be described as rarely intermittent with infrequent brief zero-flow periods, intermittent with infrequent long zero-flow periods, or highly ephemeral with frequent long zero-flow periods. However, it is clear from Fig. 2.2.7a that these subclasses are imposed upon what is, in reality, a spectrum and is a salient reminder that all classifications of flow regimes impose boundaries on continua.

Snelder et al. (2013) then explored the match between the flow-regime classifications and various environmental characteristics of the catchments of the 628 gauging stations. There were significant associations of intermittence with regional-scale climate patterns and catchment area, shape, and slope. However, these associations were only moderate, interpreted by Snelder et al. (2013) to arise because intermittence is also controlled by processes such as groundwater table fluctuations and seepage through permeable channels that operate at subcatchment scales (Section 2.2.2). This substantial fine-scale heterogeneity was also evident in the limited spatial synchronization of zero flows (Snelder et al., 2013) and is a major constraint when applying flow-regime classifications to finer-scale ecological processes for ecohydrological analyses (e.g., Leigh et al., 2016). Nonetheless, regional classifications and derivative predictive models of patterns in flow intermittence are valuable sources of information when setting environmental flows, designing regional biomonitoring programs, and predicting broad-scale changes in flow regimes likely to arise from the effects of climate change and altered patterns of human exploitation of water resources (Section 2.2.4).

Potential effects of climate change on flow regimes in IRES are attracting particular attention because of the implications for river-dependent ecosystems and availability of potable water in arid and semiarid regions. Runoff and streamflow (especially minimum flows) may begin to decrease in the future, with numbers of zero-flow days increasing in several regions (e.g., McKerchar and Schmidt, 2007; Das et al., 2011; Döll and Schmied, 2012; Jaeger et al., 2014). However, there is substantial uncertainty about how climate change will affect the frequency, occurrence, and temporal dynamics of IRES flow regimes. Not all regions of the world are expected to get drier under climate change, and there is some evidence that IRES flow regimes are fairly resilient to climate change (e.g., Botter et al., 2013; Eng et al., 2016).

In arid and semiarid western North America, climate change scenarios predict longer and drier summers with decreases in mean annual streamflow, late summer precipitation, and late summer streamflow (Seager et al., 2013). To assess the likely effects of these predicted changes on flow regimes of perennial and intermittent streams in the region, especially hydrological thresholds associated with shifts from perennial to intermittent streamflow, Reynolds et al. (2015) analyzed historic flow records from streams in the upper Colorado River Basin. The hydrological metrics of mean number of zero-flow days per year and the percent of months in the flow record that had no flow for the entire month were used to distinguish categories of flow intermittence. Stream reaches were classed as “strongly intermittent” when >5% of months over the period of record had zero flows and there were >20 zero-flow days per year on average, “weakly intermittent” when 0%–5% of months had zero flows and the number of zero-flow days averaged 1–19 days per year, and “perennial” when both the percent of zero-flow months and the average number of zero-flow days were zero.

To assess relationships between climate and flow intermittence, Reynolds et al. (2015) analyzed how the Palmer Drought Severity Index (PDSI) and annual variability in temperature and precipitation influenced the number of zero-flow days for the two intermittent stream types. Although none of the individual precipitation or temperature variables had significant explanatory power, PDSI was a significant predictor of the number of zero-flow days in both the intermittent streamflow classes. Reynolds et al. (2015) then modeled the relationship between minimum daily flow and the predictor variables of mean

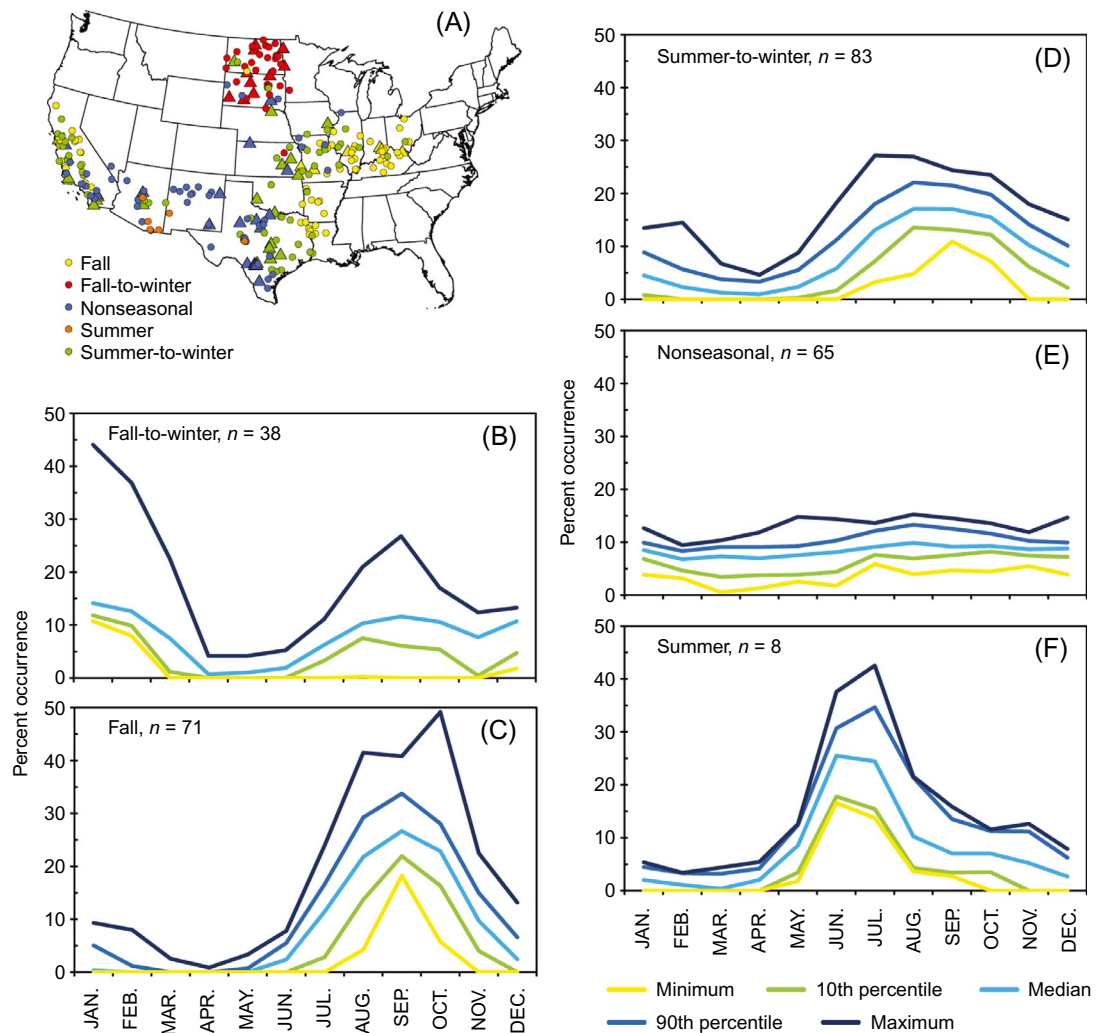
**FIG. 2.2.8**

Relationships between annual daily flow and annual minimum daily flow (A) and between zero-flow days per year and annual daily flow (B), maximum daily flow (C), and timing of maximum flow (D) in IRES of the upper Colorado River Basin. Each point represents a year's flow data from a given stream reach.

Modified from Reynolds, L.V., Shafroth, P.B., Poff, N.L., 2015. Modeled intermittency risk for small streams in the Upper Colorado River Basin under climate change. J. Hydrol. 523, 768–780, with permission.

daily flow magnitude, maximum daily magnitude, and annual timing of maximum flow for strongly and weakly intermittent streams. Annual mean daily flows were positively associated with annual minimum daily flows (Fig. 2.2.8A). There was a negative relationship between the number of zero-flow days per year and annual mean daily flow (Fig. 2.2.8B) but not maximum daily flow or its timing (Fig. 2.2.8C and D). These findings suggest that under a drier future climate, decreased mean flows will be associated with reduced minimum flows and increased number of zero-flow days, and match the predictions by Jaeger et al. (2014) that the frequency of zero-flow days would increase and flowing portions of the stream would decrease in the Verde River basin in Arizona under future climate scenarios.

A broader-scale analysis of potential climate change effects on the flow regimes of IRES across North America focused on the sensitivity of several hydrological metrics to historical variability in climate (Eng et al., 2016). To eliminate streams that seldom dry, only streams with a minimum annual average over the entire period of record of at least 15 days of zero flow per year were used. Further data screening sought only streams minimally impacted by direct human modifications and that had a minimum length of record of 10 years between 1950 and 2012, resulting in a final dataset of 256 streams. Daily flow records were used to calculate annual time series of three metrics: number of zero-flow days, average of the central 50% range of flows, and average of the largest 10% of flows. The 256 IRES were classified based on the seasonality of zero-flow periods into five different groups as fall, fall-to-winter, nonseasonal, summer, and summer-to-winter IRES (Fig. 2.2.9). Not surprisingly, many

**FIG. 2.2.9**

(A) Locations of US Geological Survey stream gauges on 265 IRES with 10–47 years of record (*circles*) and >48 years of record (*triangles*). Colors represent five broad patterns of timing when the zero-flow values occurred: fall (*yellow*), fall-to-winter (*red*), nonseasonal (*blue*), summer (*orange*), and summer-to-winter (*green*). Percent occurrence of zero flows across all gauges in each class between 1950 and 2012 for (B) fall-to-winter, (C) fall, (D) summer-to-winter, (E) nonseasonal, and (F) summer IRES.

Reproduced from Eng, K., Wolock, D. M., Dettinger, M.D., 2016. Sensitivity of intermittent streams to climate variations in the USA. *River Res. Appl.* 32, 885–895, with permission.

of the representatives of these five groups of IRES flow regimes are spatially segregated although the extent of this segregation varies (Fig. 2.2.9A); IRES with a fall-to-winter flow regime are more tightly grouped regionally than fall and summer-to-winter subclasses. One of the most interesting conclusions from this study is that future climatic changes may alter different components of the flow regimes of IRES, many of which are crucial ecological cues (Chapters 4.3, 4.5, and 4.6). This study also highlights the value of using multiple analytical approaches to flow-regime analysis, including analyses of changes to seasonal hydrographs for different hydrological metrics.

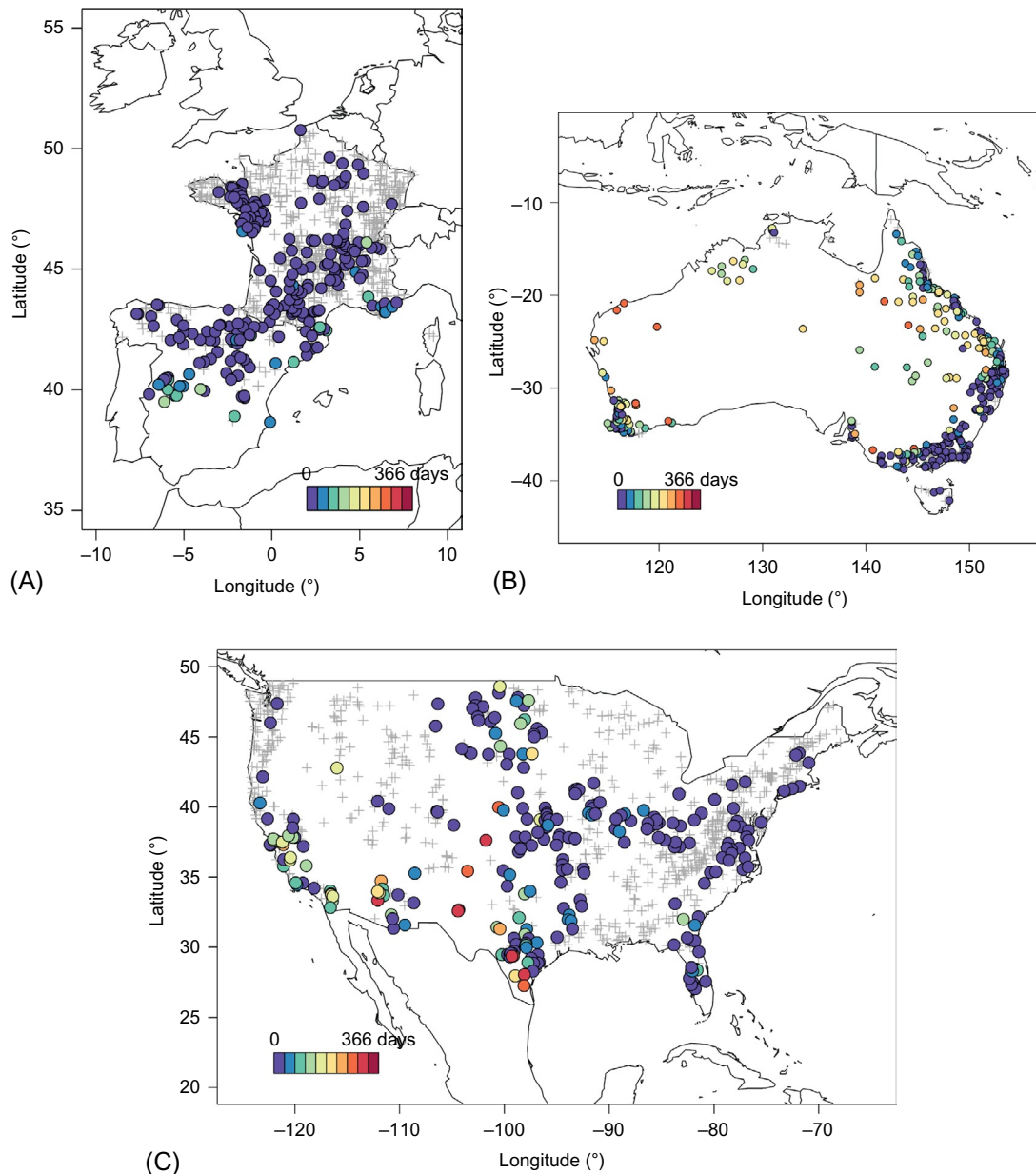
The final case study in this section is perhaps the most ambitious one, seeking to document the spatial distribution of IRES across much of the globe as well as the year-to-year variability of zero-flow events. Undertaken as part of the international research project “Intermittent River Biodiversity Analysis and Synthesis” (IRBAS, <http://irbas.cesab.org>), daily discharge data from more than 6069 gauging stations with little or no significant human influence on flow were collected from France, Spain, Australia, and the conterminous United States. Datasets comprised those with at least 30 years of records (spanning 1970–2013) and missing <5% of their data.

IRES were considered to be those averaging at least 5 zero-flow days per year, yielding a final dataset of 525 stations (Fig. 2.2.10). The proportion of IRES ranged from 8% (France) to 65% (Australia), with rivers in Australia and Spain by far the most subject to flow intermittence. Across all four countries, almost 60% of IRES experienced zero flow for at least 10% of their record length. The flow regimes of the IRES were described by two climate descriptors and 19 hydrological metrics, including 11 metrics related to the different facets of intermittence. These included the mean frequency of days without flow per year, the mean total number of zero-flow days per year, and the mean timing of zero-flow events (θ) and its dispersion. At the global scale, the most relevant explanatory factor of intermittence was an aridity index; not unexpectedly, the more arid the climate, the higher the probability of a river being intermittent.

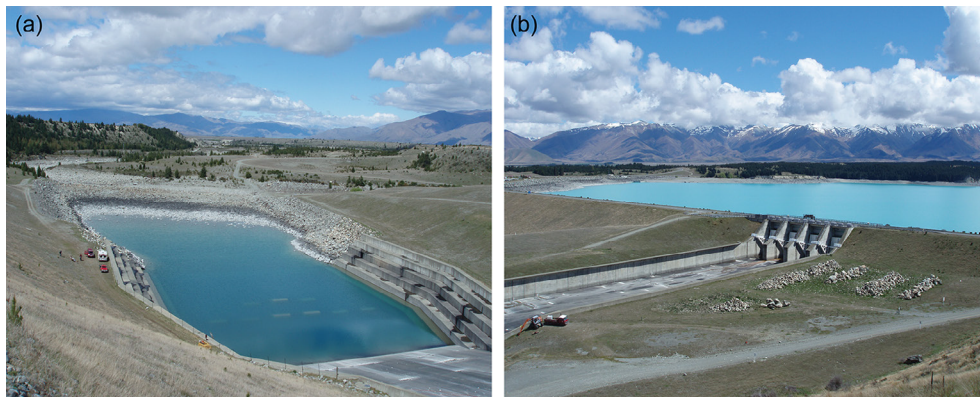
All of these case studies have substantially advanced our knowledge about the regional and global prevalence and diversity of IRES as well as the major challenges of describing their flow regimes at multiple spatial and temporal scales using various techniques and metrics. In the context of this book, these case studies and their various approaches inform the reader about hydrological aspects of flow regimes, especially intermittence, in IRES and provide the technical background for understanding how components of the flow regime (Box 2.2.1) influence physicochemical (Section 3) and ecological (Section 4) features of IRES as well as their management (Section 5). However, there is still considerable scope for further hydrological research and refinements of our perspectives about flow regimes and ecosystem processes in IRES.

2.2.5 CONCLUSIONS: RESEARCH NEEDS AND FUTURE PERSPECTIVES

The current Anthropocene era of dramatic changes to local and global hydrological cycles (Vörösmarty and Sahagian, 2000; Sterling et al., 2013) is one where IRES are increasingly prevalent across much of the globe. Many of these additional IRES may be considered “novel” ecosystems (*sensu* Hobbs et al., 2014) with altered flow regimes of increased intermittence (Fig. 2.2.11) artificially created by burgeoning demands for fresh water (Chapter 5.1). Consequently, it is now imperative that we appropriately

**FIG. 2.2.10**

Distribution of perennial rivers (*crosses*) and IRES (*colored circles*) in France and Spain (A), Australia (B), and the conterminous United States (C), expressing intermittence as the mean frequency of zero-flow days per year in equal 36-day intervals (i.e., *purple*=<36, *blue*=36–71, etc.).

**FIG. 2.2.11**

Human activities modify flow regimes to create “novel” ecosystems with increased or reduced intermittence, as seen here in the Pukaki River (a) looking downstream from the spillway of the Pukaki Dam (b), South Island, New Zealand.

Photos: Courtesy A. Boulton.

protect or manage the flow regimes of natural and modified IRES to support them as functional ecosystems (Arthington et al., 2014), potentially requiring strategies such as “reconciliation ecology” to restore severely impaired IRES (Moyle, 2014). Such protection and management often involves implementing environmental flows to supply water in the quantities and flow regimes needed to sustain natural or human-modified IRES (Acreman et al., 2014). This requires knowledge of historical, current, and likely future flow regimes of IRES, including frequency, timing, and duration of intermittence and the implications for hyporheic flow and surface water availability.

One likely future research initiative will be a far greater focus on IRES whose flow regimes have been so modified that they are either “novel” ecosystems with no current analogs or are “hybrid” combinations that retain some original characteristics but have novel elements (Hobbs et al., 2014). The most significant research gap here is understanding the relationship between components of the flow regime (e.g., duration or frequency of zero-flow periods) and the occurrence and persistence of pools and hyporheic flow in IRES and their ecosystem services (Chapter 5.2), biota (Chapters 4.1–4.6), and ecological processes (Chapters 4.7–4.9). Without this information at catchment or broader scales, restoration or protection strategies involving environmental flows (e.g., Seaman et al., 2016) are severely hampered (Chapter 5.4).

Another future initiative takes advantage of the rapid development of new technologies for gathering remotely sensed data, including time series information about the presence of surface water across the landscape, along with wet-dry mapping by citizen scientists (Section 2.2.3). With the growing realization of the increasing extent of intermittence in many parts of the world (Chapter 1), there is a need to better document “hot spots” of change in flow regimes, groundwater-surface water interactions, and catchment-level responses to hydrological impacts of human activities and climate change in IRES (Section 2.2.4). These technologies include real-time tracking of water persistence and quality in critical refugia during dry periods (King et al., 2015) so that remediation strategies can be timely and effective. The perspective that IRES may provide crucial “early warning” indicators of

landscape-scale changes in flow regimes and hydrological cycles (Palmer et al., 2008; Brooks, 2009) is likely to evolve into one that uses technological advances in mapping and flow monitoring to identify which aspects of IRES are especially sensitive, where nonlinear responses may occur in hydrological and ecological indicators to changes in flow regime, and what drivers are responsible for these changes in different IRES.

We know very little about the importance of antecedent conditions in IRES as a determinant of subsequent flow regimes and ecohydrological responses to intermittence. We are unaware of any hydrological metrics that specifically describe the rate of change in discharge before and after zero-flow periods (e.g., how fast is the change between flowing and nonflowing, and between nonflowing and flowing, periods?) yet such metrics (proposed in Table 2.2.1) are easy enough to calculate given the existence and wide use of the general rate-of-change metrics in the literature (Olden and Poff, 2003) and software such as RAP (River Analysis Package, Marsh et al., 2003). The challenge will be to decide how long before and after the zero-flow period, and which flow events, should be included in the calculations. Information across a broad range of IRES on the persistence of pools after flow cessation is scant (but see Hamilton et al., 2005) and more use could be made of citizen science (Datry et al., 2016a,b) and interviews and aerial photography (Gallart et al., 2016).

From an ecological perspective, rates of change around zero-flow periods may govern physical, chemical, and biological responses that occur during and after these events. For example, rapid resumption of flow may be associated with mass displacement of organic and inorganic materials (Corti and Datry, 2012; Rosado et al., 2015), potentially leading to poor water quality and fish kills during the dry-wet transition periods (Datry et al., 2014; King et al., 2015). Rapid loss of flow and surface water may be challenging aquatic biota more than slow loss, especially if escape mechanisms require flow (e.g., drift to downstream perennial reaches). The hydrological conditions to which biota and their habitats are exposed prior to each flow event probably determine ecological responses to zero-flow periods in IRES but currently we lack adequate empirical data from which to derive reliable generalizations.

Central to all these research needs is more detailed knowledge of the flow regimes of IRES from the short-term local scales of hourly changes in a drying pool or a riffle where flow has recently ceased through to the long-term broad scales of interannual changes across multiple catchments in landscapes presently subject to, for example, regional warming. The flow regime of all IRES is characterized by intermittence; in the next chapter, we explore how this characteristic feature affects spatial patterns of hydrological connectivity in longitudinal, lateral, and vertical dimensions. The interaction of these three spatial dimensions of connectivity with the temporal dimension of the flow regime (particularly intermittence) governs virtually all physical, chemical, and biological processes in IRES yet many hydrological aspects remain elusive and poorly understood.

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